

tests/fixtures/generate_test_pack.py

```
#!/usr/bin/env python3
"""
tests/fixtures/generate_test_pack.py
=====
Generates a synthetic CLC-structured Gateway 2 application pack for testing.

Usage:
    python tests/fixtures/generate_test_pack.py

Outputs:
    tests/fixtures/test_pack/      21 synthetic PDFs in CLC folder structure
    tests/fixtures/ground_truth.py expected analysis results for all layer tests

Deliberate coordination conflicts (embedded at known pages and sections):
    CONFLICT-001 Part A cladding load 2.5 kN/m² (2.1.1 p.7) vs 1.8 kN/m² (2.3.3 p.4)
    CONFLICT-002 Part B sprinkler temp 68 °C (2.2.1 p.12) vs 93 °C (2.2.3 p.3)

Deliberate completeness gaps (topics absent from all documents):
    GAP-001 Part A: disproportionate collapse / robustness strategy / key element
    GAP-002 Part B: cavity barrier with positional reference (grid / level / elevation)
"""

from __future__ import annotations

import os
import re
import sys
import textwrap
from pathlib import Path

from reportlab.lib import colors
from reportlab.lib.enums import TA_CENTER, TA_JUSTIFY, TA_LEFT
from reportlab.lib.pagesizes import A4
from reportlab.lib.styles import ParagraphStyle, getSampleStyleSheet
from reportlab.lib.units import cm, mm
from reportlab.platypus import (
    PageBreak,
    Paragraph,
    SimpleDocTemplate,
    Spacer,
    Table,
    TableStyle,
)

# =====
# Paths
# =====

FIXTURES_DIR = Path(__file__).parent
PACK_DIR = FIXTURES_DIR / "test_pack"
GROUND_TRUTH = FIXTURES_DIR / "ground_truth.py"

PROJECT = "Silverline House – 25-Storey Residential HRB"
PROJECT_REF = "SLH-2026"
REVISION = "Rev A"
ISSUE_DATE = "March 2026"

# =====
# Document manifest (per Guidance Note 06 / Annex 6A)
# =====

DOCUMENTS = [
    # General Application Information
    {"ref": "0.1", "title": "Application Information Schedule",
     "folder": "General_Application_Information", "part": "ADMIN"},
    {"ref": "0.2", "title": "Application Project Brief",
     "folder": "General_Application_Information", "part": "ADMIN"},
    {"ref": "0.3", "title": "Application Folder Structure and Contents Schedule",
     "folder": "General_Application_Information", "part": "ADMIN"},
    # Part A – Structure
```

```

{"ref": "2.1.1", "title": "Structural Calculation Pack",
 "folder": "Part_A_Structure", "part": "A"},
{"ref": "2.1.2", "title": "Structural General Arrangement Drawings",
 "folder": "Part_A_Structure", "part": "A"},
{"ref": "2.1.3", "title": "Geotechnical Ground Investigation Report",
 "folder": "Part_A_Structure", "part": "A"},
{"ref": "2.1.4", "title": "Pile Design and Testing Strategy",
 "folder": "Part_A_Structure", "part": "A"},
{"ref": "2.1.5", "title": "Material Specification – Concrete and Steel",
 "folder": "Part_A_Structure", "part": "A"},
{"ref": "2.1.6", "title": "Transfer Slab Design Note",
 "folder": "Part_A_Structure", "part": "A"},
# Part B – Fire Safety
{"ref": "2.2.1", "title": "Fire Strategy Report",
 "folder": "Part_B_Fire_Safety", "part": "B"},
{"ref": "2.2.2", "title": "Fire Compartmentation Drawings",
 "folder": "Part_B_Fire_Safety", "part": "B"},
{"ref": "2.2.3", "title": "MEP Sprinkler Design Specification",
 "folder": "Part_B_Fire_Safety", "part": "B"},
{"ref": "2.2.4", "title": "Smoke Extraction Strategy Note",
 "folder": "Part_B_Fire_Safety", "part": "B"},
{"ref": "2.2.5", "title": "Fire Door Schedule",
 "folder": "Part_B_Fire_Safety", "part": "B"},
{"ref": "2.2.6", "title": "External Wall Fire Performance Assessment",
 "folder": "Part_B_Fire_Safety", "part": "B"},
# Part C – Site Preparation
{"ref": "2.3.1", "title": "Rainscreen Cladding Specification",
 "folder": "Part_C_Site_Preparation", "part": "C"},
{"ref": "2.3.2", "title": "Below Ground Waterproofing Strategy",
 "folder": "Part_C_Site_Preparation", "part": "C"},
{"ref": "2.3.3", "title": "Facade Interface and Support Design",
 "folder": "Part_C_Site_Preparation", "part": "C"},
# Mandatory Documents
{"ref": "1.1", "title": "Construction Control Plan",
 "folder": "Mandatory_Documents", "part": "ADMIN"},
{"ref": "1.2", "title": "Building Regulations Compliance Statement",
 "folder": "Mandatory_Documents", "part": "ADMIN"},
{"ref": "1.3", "title": "MOR Plan",
 "folder": "Mandatory_Documents", "part": "ADMIN"},
]

```

```

# 
# Styles
# 

```

```

def _styles():
    base = getSampleStyleSheet()
    h1 = ParagraphStyle("H1", parent=base["Heading1"],
                        fontSize=14, spaceAfter=6, spaceBefore=12)
    h2 = ParagraphStyle("H2", parent=base["Heading2"],
                        fontSize=11, spaceAfter=4, spaceBefore=8)
    h3 = ParagraphStyle("H3", parent=base["Heading3"],
                        fontSize=10, spaceAfter=3, spaceBefore=6)
    body = ParagraphStyle("Body", parent=base["Normal"],
                        fontSize=10, leading=14, spaceAfter=6,
                        alignment=TA_JUSTIFY)
    cover_title = ParagraphStyle("CoverTitle", parent=base["Normal"],
                                fontSize=18, leading=22, spaceAfter=8,
                                alignment=TA_CENTER, fontName="Helvetica-Bold")
    cover_sub = ParagraphStyle("CoverSub", parent=base["Normal"],
                              fontSize=11, leading=14, spaceAfter=4,
                              alignment=TA_CENTER)
    cover_meta = ParagraphStyle("CoverMeta", parent=base["Normal"],
                              fontSize=9, leading=12,
                              alignment=TA_CENTER, textColor=colors.grey)
    return h1, h2, h3, body, cover_title, cover_sub, cover_meta

```

```

H1, H2, H3, BODY, COVER_TITLE, COVER_SUB, COVER_META = _styles()

```

```

# 
# Page header callback factory

```

```
#
def _header_cb(doc_ref: str, title: str, revision: str):
    def _draw(canvas, doc):
        canvas.saveState()
        canvas.setFont("Helvetica", 7.5)
        left = 2.2 * cm
        right = A4[0] - 2.2 * cm
        y = A4[1] - 1.3 * cm
        canvas.drawString(left, y, f"{PROJECT_REF} | {doc_ref} | {revision}")
        canvas.drawRightString(right, y, title[:60])
        canvas.setLineWidth(0.3)
        canvas.line(left, y - 2 * mm, right, y - 2 * mm)
        canvas.setFont("Helvetica", 7)
        canvas.drawCentredString(A4[0] / 2, 1.2 * cm,
                                f"Page {doc.page}")
        canvas.restoreState()
    return _draw

#
# Filename helper
#

def _filename(doc: dict) -> str:
    safe = re.sub(r"[^a-zA-Z0-9]+", "_", doc["title"])
    safe = safe.strip("_")
    return f"{doc['ref']}_{safe}_RevA.pdf"

#
# Core PDF builder
#

def _build_pdf(path: Path, doc_ref: str, title: str, pages: list[list]):
    """
    Write a PDF. `pages` is a list of lists-of-flowables; each inner list
    becomes exactly one page (a PageBreak is appended between them).
    """
    story = []
    for i, page_content in enumerate(pages):
        story.extend(page_content)
        if i < len(pages) - 1:
            story.append(PageBreak())

    cb = _header_cb(doc_ref, title, REVISION)
    doc = SimpleDocTemplate(
        str(path),
        pagesize=A4,
        topMargin=2.0 * cm,
        bottomMargin=2.0 * cm,
        leftMargin=2.2 * cm,
        rightMargin=2.2 * cm,
    )
    doc.build(story, onFirstPage=cb, onLaterPages=cb)

#
# Cover page builder
#

def _cover(doc_ref: str, title: str, subtitle: str = "") -> list:
    items = [
        Spacer(1, 3 * cm),
        Paragraph(PROJECT, COVER_SUB),
        Spacer(1, 0.5 * cm),
        Paragraph(title, COVER_TITLE),
    ]
    if subtitle:
        items.append(Paragraph(subtitle, COVER_SUB))
    items += [
        Spacer(1, 1 * cm),
    ]
```


[illegible]

"The evacuation strategy for Silverline House adopts a simultaneous evacuation policy "
"for the residential floors. Each apartment has a front door opening onto a protected "
"corridor with a maximum travel distance to the nearest staircase of 7.5 m. The two "
"protected staircases are located at opposite ends of the building core, more than "
"15 m apart in plan, ensuring they cannot be simultaneously compromised by a single "
"fire event. Evacuation timing analysis carried out in accordance with PD 7974-6 "
"demonstrates that all occupants can reach a place of relative safety within "
"3.5 minutes of alarm activation, including 30 seconds pre-travel time. Refuges are "
"provided at each staircase on every residential floor for persons who are unable to "
"use the stairs independently. The premises information box is located in the ground- "
"floor lobby adjacent to the firefighting lift lobby."

)

NOTE: cavity barriers mentioned here only generically – no grid/level/elevation refs

_COMPARTMENTATION_TEXT = (
"The building is divided into fire compartments in accordance with Approved Document B "
"Volume 1, Table B1. Each apartment forms a separate fire compartment. The protected "
"corridors are enclosed by fire-resisting construction providing at least 30 minutes "
"fire resistance. The staircase enclosures provide 60 minutes fire resistance. "
"Cavity barriers are provided within the external wall construction in accordance "
"with the requirements of Approved Document B to limit the spread of fire through "
"concealed spaces. The cavity barrier specification and extent of installation will "
"be confirmed in the Fire Compartmentation Drawings. Service penetrations through "
"compartment walls and floors are sealed with intumescent collars or fire-stopping "
"products with third-party certification to the required fire resistance period."

)

_COMPARTMENT_WALLS_TEXT = (
"Compartment walls between apartments are constructed in 200 mm dense aggregate "
"blockwork or equivalent lightweight block achieving 60 minutes fire resistance in "
"accordance with BS EN 1996-1-2 and the tabulated data in Approved Document B. "
"Where the compartment wall coincides with the structural frame, the wall abuts the "
"underside of the floor slab above and is sealed to prevent smoke passage. "
"Junctions between compartment walls and the external facade are detailed to maintain "
"compartmentation integrity at the facade interface. Product certificates and fire "
"test evidence for all specified masonry units are included in the Construction "
"Product Schedule accompanying this application."

)

_FLOOR_CEILING_TEXT = (
"Compartment floors are formed by the reinforced concrete flat slab construction "
"(minimum 200 mm depth), which inherently provides 120 minutes fire resistance "
"when designed in accordance with BS EN 1992-1-2. Where a suspended ceiling is "
"provided below the structural slab, the ceiling system has been designed to "
"maintain the compartment floor fire resistance rating. Penetrations through "
"compartment floors for services (drainage, ventilation ducts, electrical conduits) "
"are sealed with proprietary fire-stopping systems carrying third-party certification "
"to BS EN 1366-3. The specification for these systems is contained in the Services "
"Fire Stopping Schedule."

)

_DETECTION_TEXT = (
"An automatic fire detection and alarm system has been designed in accordance with "
"BS 5839-1 Category L2 for the common areas and Category LD2 for the individual "
"apartments. Optical smoke detectors are installed in all protected corridors, "
"lobbies, and staircase enclosures. Combination heat/smoke detectors are installed "
"in apartment entrance halls. The system is connected to a central addressable "
"control panel located in the ground-floor security room with a sub-panel in the "
"Level 13 plant room. Manual call points are installed at each staircase landing "
"on every floor and at each exit door at ground level. The detection system is "
"interfaced with the building management system (BMS) to allow integration with "
"smoke control, lift recall, and door hold-open release functions."

)

_AUTO_DETECTION_TEXT = (
"Apartment detection comprises addressable multi-sensor detectors (optical/heat) "
"in the entrance hall, interconnected with a standalone sounder within the apartment "
"to provide early warning to occupants. The detection system is designed to initiate "
"the common area alarm on detection of fire within any apartment, triggering "
"simultaneous evacuation. Detectors are positioned at 150 mm from the ceiling "
"surface in accordance with BS 5839-1 Table B1. All detectors are accessible from "
"floor level using a standard detector exchange tool without requiring scaffolding "

[illegible]

[illegible]

[Paragraph("3 Evacuation Strategy", H2), Spacer(1, 0.3 * cm),
Paragraph(_EVACUATION_TEXT, BODY), Spacer(1, 0.2 * cm),
Paragraph(
"Cavity barriers are to be provided throughout the facade construction in "
"compliance with Approved Document B. Full details of cavity barrier "
"specification and installation will be confirmed in the Fire "
"Compartmentation Drawings.",
BODY)],

Page 5

[Paragraph("4 Fire Compartmentation", H2), Spacer(1, 0.3 * cm),
Paragraph(_COMPARTMENTATION_TEXT, BODY), Spacer(1, 0.2 * cm),
Paragraph(_STANDARDS_TEXT, BODY)],

Page 6

[Paragraph("4.1 Compartment Walls", H2), Spacer(1, 0.3 * cm),
Paragraph(_COMPARTMENT_WALLS_TEXT, BODY), Spacer(1, 0.2 * cm),
Paragraph(_FIRE_INTRO, BODY)],

Page 7

[Paragraph("4.2 Compartment Floors and Ceiling Assemblies", H2),
Spacer(1, 0.3 * cm),
Paragraph(_FLOOR_CEILING_TEXT, BODY), Spacer(1, 0.2 * cm),
Paragraph(_COMPARTMENTATION_TEXT, BODY)],

Page 8

[Paragraph("5 Fire Detection and Alarm System", H2), Spacer(1, 0.3 * cm),
Paragraph(_DETECTION_TEXT, BODY), Spacer(1, 0.2 * cm),
Paragraph(_STANDARDS_TEXT, BODY)],

Page 9

[Paragraph("5.1 Automatic Fire Detection", H2), Spacer(1, 0.3 * cm),
Paragraph(_AUTO_DETECTION_TEXT, BODY), Spacer(1, 0.2 * cm),
Paragraph(_FIRE_INTRO, BODY)],

Page 10

[Paragraph("5.2 Fire Alarm System", H2), Spacer(1, 0.3 * cm),
Paragraph(_ALARM_SYSTEM_TEXT, BODY), Spacer(1, 0.2 * cm),
Paragraph(_DETECTION_TEXT, BODY)],

Page 11

[Paragraph("6 Active Fire Suppression", H2), Spacer(1, 0.3 * cm),
Paragraph(_SUPPRESSION_INTRO_TEXT, BODY), Spacer(1, 0.2 * cm),
Paragraph("6.2 Wet Riser System", H3), Spacer(1, 0.2 * cm),
Paragraph(_DRY_WET_RISER_TEXT, BODY)],

Page 12 – CONFLICT: 68°C

[Paragraph("6.3 Sprinkler System", H2), Spacer(1, 0.3 * cm),
Paragraph(
"A wet pipe sprinkler system complying with BS EN 12845:2015 is installed "
"throughout all residential floors, the retail unit, all basement car parking "
"levels, and plant rooms. The system is classified as Ordinary Hazard Group 2 "
"for the residential areas and Extra Hazard Group 1 for the basement car park. "
"Design density and area of operation comply with BS EN 12845 Table 5.",
BODY),
Spacer(1, 0.2 * cm),
Paragraph(
"The sprinkler system is designed using standard response heads with an "
"activation temperature of 68\00b0C in accordance with BS EN 12845:2015 "
"Table 4. This activation temperature is appropriate for the ambient "
"conditions in the residential areas and plant rooms. The sprinkler heads "
"comply with BS EN 12259-1 and are from an LPCB-certificated manufacturer.",
BODY),
Spacer(1, 0.2 * cm),
Paragraph(
"The water supply to the sprinkler system is from a 100,000 litre break tank "
"in the basement plant room, pressurised by a duty-and-standby pump set. The "
"pump arrangement provides a minimum flow rate of 1,800 l/min at 3.5 bar at "
"the highest sprinkler head. The full sprinkler system design, including "
"hydraulic calculations, is contained in the MEP Sprinkler Design "
"Specification (document ref 2.2.3).",
BODY)],

Page 13

[Paragraph("7 Fire Service Access", H2), Spacer(1, 0.3 * cm),
Paragraph(
"Fire service access is provided via a perimeter access road capable of "
"supporting a fire appliance with a maximum axle load of 16.3 tonnes. "
"The access road provides a clear width of 3.7 m and a clear height of "
"3.7 m under all overhead obstructions. Two firefighting lifts serving all "
"floors are provided in accordance with BS EN 81-72:2020 and Approved "

[illegible]


```
# Generate all PDFs
#
def _generate_pdfs() -> list[dict]:
    """Create all folders and write all PDFs. Returns manifest with file paths."""
    folders = {d["folder"] for d in DOCUMENTS}
    for folder in folders:
        (PACK_DIR / folder).mkdir(parents=True, exist_ok=True)

    manifest = []
    for doc in DOCUMENTS:
        fname = _filename(doc)
        path = PACK_DIR / doc["folder"] / fname
        pages = _BUILDERS[doc["ref"]]()
        _build_pdf(path, doc["ref"], doc["title"], pages)
        manifest.append(**doc, "filename": fname, "path": str(path))
        print(f" {doc['ref']:<8} {fname}")

    return manifest

#
# Write ground_truth.py
#
_GROUND_TRUTH_TEMPLATE = '''\
"""
tests/fixtures/ground_truth.py - auto-generated by generate_test_pack.py
DO NOT EDIT MANUALLY.

Expected analysis results for the synthetic test pack.
All layer tests import GROUND_TRUTH and assert against its values.
"""

GROUND_TRUTH = {{
    "total_documents": {total_documents},
    "parts_covered": ["A", "B", "C"],

    # ■■ Coordination conflicts
    # Both embedded at exact page/section in the named documents.
    "coordination_conflicts": [
        {{
            "id": "CONFLICT-001",
            "type": "NUMERICAL",
            "part": "A",
            "pattern": "cladding_load",
            "doc_a": {{
                "ref": "2.1.1", "page": 7,
                "section": "4.2 Fa\u00e7ade Loading",
                "value": 2.5, "unit": "kN/m2"
            }},
            "doc_b": {{
                "ref": "2.3.3", "page": 4,
                "section": "3.1 System Self-Weight",
                "value": 1.8, "unit": "kN/m2"
            }},
            "expected_severity": "HIGH",
        }},
        {{
            "id": "CONFLICT-002",
            "type": "NUMERICAL",
            "part": "B",
            "pattern": "sprinkler_activation_temperature",
            "doc_a": {{
                "ref": "2.2.1", "page": 12,
                "section": "6.3 Sprinkler System",
                "value": 68, "unit": "\u00b0C"
            }},
            "doc_b": {{
                "ref": "2.2.3", "page": 3,
                "section": "2.1 Head Specification",
                "value": 93, "unit": "\u00b0C"
            }}
        }}
    ]
}}
```



```
f'filename': {d["filename"]!r}}},'
```

```
)
documents_repr = "[\n" + "\n".join(doc_lines) + "\n    ]"

content = _GROUND_TRUTH_TEMPLATE.format(
    total_documents=len(manifest),
    documents_repr=documents_repr,
)
GROUND_TRUTH.write_text(content, encoding="utf-8")
print(f"\n ground_truth.py ({len(manifest)} documents)")

# #####
# Verification – check forbidden phrases are absent
# #####

def _verify_gaps(manifest: list[dict]) -> None:
    """
    Post-generation sanity check using pdftotext if available.
    Warns (does not fail) if forbidden phrases are found.
    """
    try:
        import subprocess
        pdftotext = "pdftotext"
        subprocess.run([pdftotext, "--version"], capture_output=True, check=True)
    except (FileNotFoundError, subprocess.CalledProcessError):
        print("\n [skip] pdftotext not available – gap verification skipped")
        return

    import subprocess

    forbidden_a = ["disproportionate collapse", "robustness strategy", "key element"]

    part_a_docs = [d for d in manifest if d["part"] == "A"]
    for doc in part_a_docs:
        result = subprocess.run(
            [pdftotext, doc["path"], "-"], capture_output=True, text=True
        )
        text = result.stdout.lower()
        for phrase in forbidden_a:
            if phrase in text:
                print(f" [WARN] GAP-001 VIOLATED: '{phrase}' found in {doc['ref']} {doc['filename']}")

    part_b_docs = [d for d in manifest if d["part"] == "B"]
    import re
    pattern = re.compile(r"cavity barrier.{0,80}(?:grid|level|elevation|shown on|position)", re.IGNORECASE)
    for doc in part_b_docs:
        result = subprocess.run(
            [pdftotext, doc["path"], "-"], capture_output=True, text=True
        )
        if pattern.search(result.stdout):
            print(f" [WARN] GAP-002 VIOLATED: positional cavity barrier reference in {doc['ref']} {doc['filename']}")

    print(" Verification complete.")

# #####
# Entry point
# #####

def main() -> None:
    print(f"Generating test pack → {PACK_DIR}")
    print(f"Project: {PROJECT}\n")

    manifest = _generate_pdfs()
    _write_ground_truth(manifest)
    _verify_gaps(manifest)

    print(f"\nDone – {len(manifest)} PDFs written to {PACK_DIR}")
    print(f"Ground truth → {GROUND_TRUTH}")

if __name__ == "__main__":
    main()
```